The role of ISDN in data networking

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The integrated services digital network (ISDN) offers on-demand switched end-to-end digital connectivity over the wide area, enabling the integration of both voice and data services over a common core network. This paper considers the role of ISDN in data networking, both as the core transport network and as a means of enhancing resilience in a mixed-technology data solution, complementing other data network technologies. The issues associated with data transport based on an ISDN solution are considered, and include security, bandwidth utilisation, scalability and the management of ISDN-attached devices. Optimising the use of ISDN networks for supporting the most prevalent routed and routeing protocols is also discussed.

1. Introduction

The integrated services digital network (ISDN) offers on-demand switched digital end-to-end connectivity over the wide area. This enables the integration of both voice and data services over a common core network. This is distinct from the public switched telephone network (PSTN) which offers only analogue connectivity, requiring the use of modems for the transport of digital information.

There are currently two alternative interfaces to the ISDN network, known as the basic rate interface (BRI) and primary rate interface (PRI). The former provides two B-channels of 64 kbit/s and a single D-channel at 16 kbit/s, whereas the PRI offers any number of B-channels up to thirty and a single D-channel at 64 kbit/s. The B-channels are circuit-switched point-to-point connections, and can carry pulse code modulated (PCM) encoded digital voice, or data. The D-channel is primarily used for signalling information (i.e. out-of-band call control of the B-channels). The BRI is the more widely used interface type.

ISDN holds a unique position compared to other network services used for data transport. It is a dial-on-demand service, so enabling bandwidth on demand. There are mechanisms in place for aggregating multiple B-channels (discussed in section 4), to provide the means for network scala-bility. This is distinct from virtually all other data network-ing core technologies, which generally offer permanent con-nectivity at a fixed maximum bandwidth, e.g. BT's Switched Multi-megabit Data Service (SMDS). The two notable exceptions to this at present are the public switched telephone network (PSTN) with modems (enabling lower speed dial-on-demand), and BT's flexible bandwidth service (FBS) which offers 'scheduled' additional bandwidth. It should be noted, however, that FBS only supports permanent point-to-point connections.

A further distinguishing feature of ISDN is service pricing, in particular usage-based billing. Similar to the PSTN, ISDN pricing includes a line rental and charge related to connection time and distance (no account is taken of link utilisation). Where connection types are permanent (e.g. SMDS), pricing is generally fixed and not related to usage. Usage-based billing therefore contributes significantly to ISDN's unique position within the range of core network technologies.

This paper introduces ISDN-attached router networks, with ISDN being used in four distinct roles:

- as the core network service,
- for teleworking,
- as a back-up technology to other data services,
- for the management of routers.

A number of developments in ISDN networking to support these uses are discussed. The transport of a number of prevalent routeing protocols are introduced, along with related issues, including optimising link bandwidth, security and scalability. Finally, an example network is discussed which illustrates a typical ISDN networking solution.

The Broadband and Data Networks unit at BT Laboratories (BTL) has extensive experience in the design of advanced internetworks using ISDN and multi-protocol routers from Cisco Sytems.

2. ISDN router networks

To understand the issues associated with ISDN data networking, a basic understanding of routed and routeing protocols is required. An introduction to these can be found in King [1].

Since its launch by BT in the UK, the demand for ISDN has steadily increased; during the last year for which statistics are publicly available (1996) the total number of channels increased by over 50%. The total number of channels currently in use exceeds one million. Its increase in popularity has been largely attributable to its role within the data networking world. Within this environment, ISDN may be favoured for a number of roles — these include its use as a core network technology (this may involve teleworking), as a back-up network to ensure connectivity in the event of failure of the primary network, for dial access to both corporate intranets and the Internet, and to supplement fixed-link technologies.

ISDN as a core network technology involves making a data connection ‘on-demand’, transferring data, and clearing down the connection. The usage-based tariffing structure of ISDN encourages connection time to be kept to a minimum. This method of exchanging data is referred to as dial-on-demand. Teleworking also utilises this method of connecting, with the ISDN connection generally being made direct to a card within the portable PC itself. Connection is usually made between the teleworker and a central (host) site.

ISDN may be used in conjunction with fixed-link network technologies, such as Frame Relay, in order to provide resilience to the fixed-link network in the form of a back-up link; using ISDN in this way is referred to as dial back-up. In addition, a technique known as dial top-up allows ISDN connections to supplement fixed-link connections to other core networks, such as leased line, to provide additional bandwidth during peak periods.

Apart from teleworking, each of the ISDN roles discussed involves collocating a router at each of the connected sites. These roles are now discussed.

2.1 Dial on-demand

Dial on-demand enables routers and other ISDN customer premises equipment (CPE) to exchange data over an ISDN network in an efficient way, with ISDN connections only being established when an exchange of data is required.

Taking the example in Fig 1, an individual teleworker and a branch site require access to a central host. In this example, the teleworker connects to the host site using dial on-demand across ISDN at various times throughout the day to collect e-mail and transfer files. The branch site uses dial on-demand to back up its data to the host site at the end of each working day. Neither the teleworker nor the branch site are permanently connected to the host site, and the connection is cleared shortly after data transfer is complete.

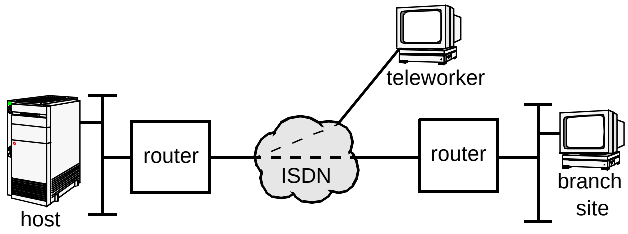

Fig 1 Example network for dial on-demand ISDN.

Dial-on-demand using ISDN is ideally suited to applications requiring occasional data transfer only. At the current time, if communication for an aggregate of more than 3 to 4 hours per day is required, a fixed-link technology, such as KiloStream, may be a better option than ISDN; this may not be appropriate to a (mobile) teleworker. For ISDN connectivity to be used effectively, connection times should be carefully traded off against the need for up-to-date network information, e.g. do not continually make a connection to pass routeing updates (see section 5). This trade-off is ultimately the choice of the customer.

2.2 Dial back-up

ISDN dial back-up is used to provide resilience for a separate core networking technology, e.g. leased line (an example of which would be BT’s KiloStream service).

Using the example in Fig 2, under normal operating conditions data transfer between the host and branch sites is across the leased-line network; no ISDN connections are used. However, in the event of the leased line network connection failing, an ISDN connection is automatically established between the branch and host site; traffic is then immediately re-routed over this link. Once the leased line connection is re-established, traffic again uses this preferential path and the ISDN connection is cleared down.

With complex wide-area networking technologies, such as Frame Relay, the most commonly used method of detecting a primary network failure is by using a dynamic routeing protocol. The loss of a route across the primary network can then be easily and quickly detected (by loss of a routeing update) and the ISDN back-up link thus activated.

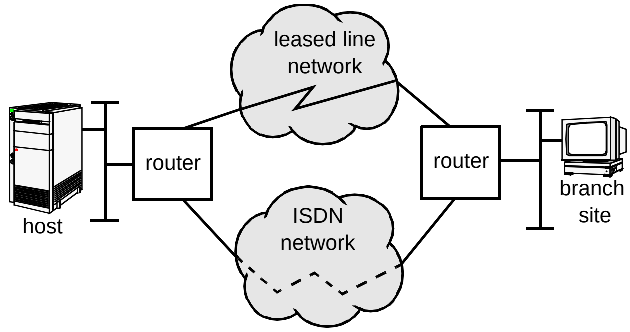

Fig 2 Example of using ISDN in conjunction with leased line.

As an ISDN call only need be established when the primary network has failed, ISDN is particularly suitable and cost effective as a back-up technology.

2.3 Dial top-up

Dial top-up can provide an efficient means of providing additional bandwidth-on-demand in certain scenarios. For example, a simple leased line connection may be supplemented by an ISDN connection during peak periods to cope with the additional traffic demands made at that time. The network designer should aim to distribute these traffic demands equally between the two links.

For example, in Fig 2 a permanent leased line link connects the branch site to the host site; for the majority of the time this link provides sufficient bandwidth for the demands made between the two sites. However, during periods of higher demand (such as large file transfers) the throughput provided by the leased line link is insufficient. To alleviate congestion during such periods, a temporary ISDN connection is established between the branch and host site, allowing data exchange using both the leased line and ISDN network.

Dial top-up combines fixed-link technology with the benefits of ISDN's bandwidth-on-demand ability to provide a more flexible networking solution.

3. Interoperability

CPE from different vendors must interoperate correctly for communication over ISDN to take place — hence adherence to standards, both at the signalling layer and the data link layer is essential. Issues pertaining to signalling layer interoperability are particularly relevant in the UK, where the original ISDN 30 service used the DASS2 signalling protocol; however, most vendors of ISDN 30 compatible equipment adhered to the international standard I.421 [2], which is not compatible with the DASS2 signalling protocol. Hence BT now offer an ISDN 30 service using the I.421 signalling protocol.

Signalling protocol compatibility between CPE ensures that when a circuit-switched call is established the user is provided with one or more transparent 64 kbit/s connections. End-user protocols must interoperate over these paths for end-to-end communication to take place; namely, protocols at the data-link layer [3] and above must interoperate.

The point-to-point protocol (PPP) [4] was developed as a standard for the interoperability of different vendors' equipment at the data-link layer. PPP is a connectionless protocol and is recommended for the interconnection of CPE, as it provides a standardised method for transporting multi-protocol datagrams. In addition, it supports a number of authentication protocols, the key two being the password authentication protocol (PAP) and challenge handshake authentication protocol (CHAP) [5].

4. Bandwidth optimisation

Tariffs for ISDN have a connection time component, with the consequence that network design and router configuration are generally tuned to minimise connection time and latency, while maximising the data throughput. In addition, certain users and/or applications may require a higher throughput than that which can be provided by a single ISDN B-channel (i.e. greater than 64 kbit/s). In order to optimise network design to meet these individual requirements there are a number of options available, which are now discussed.

4.1 Bandwidth-on-demand

Consider the case of an ISDN-only router network, with a central host and a number of branch sites requiring access to the host. In this case the requirement may be for more than 64 kbit/s bandwidth at peak times, i.e. greater than a single B-channel can provide. This is illustrated in Fig 3.

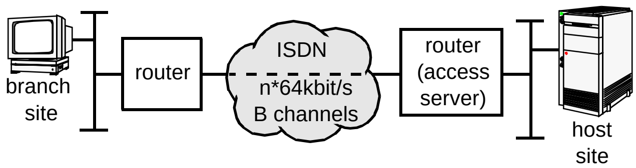

Fig 3 ISDN bandwidth-on-demand.

The interface between the router(s) and the ISDN network may be one or more BRIs or PRIs. Increased bandwidth is achievable by initiating additional B-channel connections, based on preconfigured load thresholds being reached, and aggregating the traffic they carry. There are protocols defined to enable this, the most widely deployed being the multilink point-to-point protocol (MP) [6]. This is an extension of the PPP protocol [5]; it provides a mechanism for forwarding packets across a number of parallel PPP circuits to the same destination. The protocol ensures that packet fragments are sequenced; hence packet ordering is maintained at the destination router for onward forwarding to the end station. Multilink is negotiated during the initial PPP link control protocol option phase. MP enables multiple links from the same client router to terminate at the **same** destination router (often referred to as an access server).

Thereby, the aggregated B-channels may be treated as a single higher-bandwidth bit pipe between the communicating routers. Many B-channels on multiple interfaces may be aggregated in this way, offering n*64 kbit/s dial-on-demand bandwidth. MP will entail a higher processing overhead on the router.

A more recent development, primarily aimed at making ISDN solutions more scalable, is Cisco's Multichassis Multilink PPP (MMP). This is a development of MP, and enables a single site using several ISDN connections to be terminated across multiple ISDN interfaces on more than one router. The latter routers, often referred to as 'access routers', are generally located at a host site and must be connected via a common backbone (e.g. Ethernet). All received traffic will be forwarded to one of the routers, which acts as the master for that particular user session, thus performing the same functionality as MP. This is illustrated in Fig 4.

Typically, the ISDN calling line ID — effectively the telephone number — at the host site will form part of a hunt group, i.e. where a single ISDN number is allocated to more than one ISDN line. Each line is used in a round-robin (sequential) fashion. An added advantage of a hunt group is resilience; in the event of a single line failure, connectivity is still possible using an alternative ISDN line. In the example shown in Fig 5, routers C, D and E would all share the same ISDN number even though they are connected by separate physical interfaces to the ISDN itself.

User A (which could be a branch site) dials into the host site via the ISDN. The first connection is to router C, which will become the 'master' for this user session by default, i.e. the 'owner' of this connection bundle. Hence a second call from user A, although answered by router D, will be forwarded via the backbone Ethernet to router C, as C has declared itself 'master' for user A. PPP de-encapsulation for **both** PPP connections will occur in router C, which will subsequently forward the de-encapsulated re-sequenced traffic to the host. In the reverse direction router C will load share the traffic between the two links.

Similarly, User B dials into the host site via the ISDN. In the example shown in Fig 4, two connections are made, one terminating at router D, one at router E. Assuming the first connection was to router D, it will thus become 'master' of this connection bundle. Were the first connection to have come into router C, it would by default be the 'master' for the user B bundle as well; this places a significant processing overhead on the router. One method of reducing this overhead is by introducing an 'offload server' — this is generally a router dedicated to being the 'master' for all bundles, and typically will not be ISDN-attached. This then relieves the processing overhead on the ISDN-attached routers. It should be noted that the router receiving the initial connection for a bundle need not become the master; this is manually configurable (if desired) by altering a 'seed-bid' metric.

4.2 *Prioritisation of traffic*

Prioritisation and queuing of traffic may be employed when there is congestion across the WAN (including serial lines and ISDN), due to bandwidth being at a high premium. This is frequently the case over the wide area, in particular during peak periods. Prioritisation enables the router to treat particular traffic types differently, by queuing packets

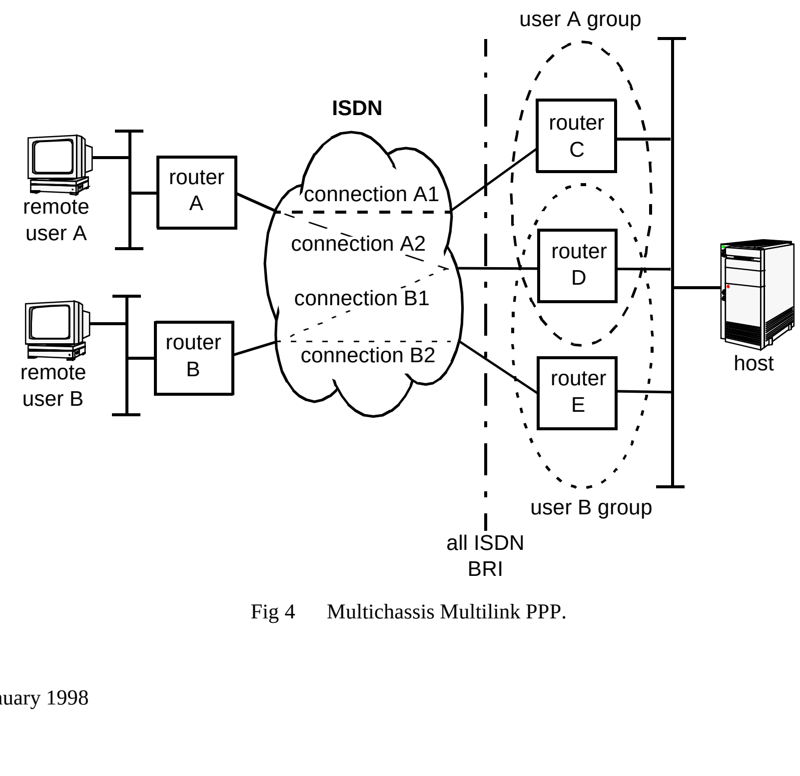

Fig 4 Multichassis Multilink PPP.

appropriately. For example, IP e-mail [7] may be prioritised over all other IP traffic, or more generally all IP traffic may be prioritised over IPX and Appletalk. The most significant disadvantage of prioritisation and queuing is that they introduce latency, as further processing and buffering are required in the router, and typically throughput is also reduced as a result. Thus, prioritisation should not be used where bandwidth is not at a high premium, e.g. in a LAN type environment. It should be noted that new switching techniques have been developed to offset this overhead. (The techniques described here are those offered by Cisco, the world's largest router manufacturer.)

There are three main techniques for the prioritisation of traffic.

- Priority queuing

  This prioritises traffic based on traffic type (e.g. IP, IPX, Appletalk) and places it in an appropriate queue (note that TCP/IP may prioritise on port numbers). The highest priority queue is serviced first until it is empty, after which the next priority queue is serviced, and so on.

- Customised queuing

  This guarantees some level of service to all traffic. Different traffic types are assigned particular queue depths, and service is on a round-robin basis. Therefore a specific protocol is prioritised by assigning it more queue space, but the protocol will never monopolise the entire bandwidth.

- Fair queuing

  This generally prioritises traffic according to conversation (i.e. application) type. One such algorithm, Cisco's Weighted Fair Queuing (WFQ), dynamically sorts traffic into messages that are part of a conversation or flow (typically based on source address, destination address and application type). The messages queued for low-volume conversations, usually interactive traffic, are given priority over bandwidth-intensive conversations, such as file transfers.

The choice of queuing method is ultimately based on the application(s) in use and the bandwidth available. Priority queuing is used to ensure critical traffic gets through, generally with low-bandwidth links. When more bandwidth is available, custom queuing equitably shares between the different traffic types, and is best suited to high-bandwidth links. The choice ultimately depends on the user's own specific requirements.

4.3 Compression

Compression may be used to reduce the actual payload carried by a connection, without the loss of any information. The result of compression is to increase the effective throughput of a particular connection. Compression entails a significant processing overhead within the device providing it, often the router. Careful consideration must therefore be given to user traffic profiles before employing compression, as other tasks (in particular routeing within routers) may be adversely affected.

As there are a number of *de facto* algorithms in use (e.g. Predictor, Stacker), the network administrator must ensure commonality at the ends of the link. A compression ratio of 2:1 is typical for a document transfer where the document contains a mix of text and diagrams. Compression is frequently used in ISDN networks where only a limited number of B-channels are used because there is a definite impact on router CPU utilisation. However, customers are not usually prepared to pay for more CPU power in the router, as compression on the desktop (e.g. zip files) is cheaper to buy, even though this may not be an efficient, cost-effective solution.

4.4 Further considerations

Any network traffic not carrying user data may be considered an overhead, consuming valuable network bandwidth, and should be minimised where possible. This is of particular relevance to ISDN, where bandwidth is generally at such a high premium. Such traffic could include dynamic routeing updates, protocol session 'keepalives' and protocol service advertising, depending on the routeing method and transport protocols in use. Various methods for minimising this type of traffic are discussed in section 5, with solutions employing static routes, spoofing (see section 5.3), and static server configuration.

It should be noted that while bandwidth optimisation using techniques such as compression and prioritisation may be employed, these will entail a processing overhead within the router and thus impact on the router's throughput and performance.

5. Protocol issues

The use of ISDN for data networking purposes has increased dramatically in both domestic and business environments. Businesses in particular are making use of ISDN for data transfer in the scenarios discussed in the previous sections. Any data exchange or communication will require a choice of data protocol, i.e. the agreed way in which end-stations will communicate; this is analogous to human beings needing to speak the same language in order to communicate. Hence, before network managers and administrators begin using ISDN as the primary means of transport for their mission-critical data, there is a wide range of data protocol issues which must be considered. This section provides an insight into the issues associated with a number of the more common data protocols, and the impact of these on an ISDN network. Current solutions for optimising the use of these protocols across the ISDN are then discussed.

Routed and routeing protocols were initially designed for use over wide area networks (WANs) using fixed links; fixed bandwidth was available at all times and customers paid a flat tariff for the use of this service. Thus, routed and routeing protocols were designed with fixed links in mind and have therefore tended to be very ‘chatty’; this presents a number of challenges when used across ISDN dial-up networks, as certain routed protocols send large amounts of administrative traffic for end-to-end user sessions to take place. This administrative traffic may incur significant call charges when used across ISDN networks, as calls will either be initiated or maintained each time the administrative traffic is to be forwarded. Several of the design methods to overcome this are now described.

5.1 Administrative traffic across dial-up — issues

The types of administrative traffic can be categorised into certain areas.

- Routeing information

  As described in King [1], routeing updates advertise the reachability of destination networks and are split into two categories — distance-vector and link-state protocols. Distance-vector protocols (e.g. IP RIP [8, 9], IPX RIP [10]) transmit periodic updates, whereas link-state protocols (e.g. OSPF [11]) transmit updates when there is a topology change. Both periodic and reactive routeing updates can cause an ISDN B-channel to be initiated or maintained, depending on the customer’s requirements. Increasing periodic routeing update timers may reduce ISDN call charges (as ISDN calls are made less frequently); however, there will be a trade-off in ‘convergence’ time (the time taken for new routeing information to be propagated throughout the network and establish a consistent set of new routeing tables).

  It should be noted that generally router configuration is such that dynamic routeing information does not cause the initiation or maintenance of the ISDN connection; while the ISDN connection is down then static routeing is used (i.e. non-dynamic and manually configured). However, once the ISDN connection is initiated, dynamic routeing updates are passed across the ISDN connection and replace any identical static routes within the router. Following the clearing of the connection, the dynamic routeing entries will time out and consequently be replaced in the router by identically configured static routes. Static routes used in this way are referred to as floating static routes, and are used extensively in ISDN data networking.

- Service information

  Novell NetWare is one of the more popular client/server architectures used in both small and large corporate networks. In addition to a dynamic distance vector routeing protocol (IPX RIP), Novell NetWare makes use of a service advertisement protocol (SAP). The SAP allows service-providing nodes such as file servers and print servers to advertise their resources throughout the network. These advertisements are sent periodically and allow clients within the network to select/access the appropriate required services. Figure 5 depicts an IPX file server forwarding periodic SAP updates throughout the network. Further information on IPX routeing and IPX service updates is available in the specification [10].

  Dynamic SAP updates will cause an ISDN B-channel to be initiated or maintained, increasing a customer’s total connection time. Increasing the periodic SAP update timers will reduce ISDN connection time; however, this may lead to service instability if particular servers become unavailable.

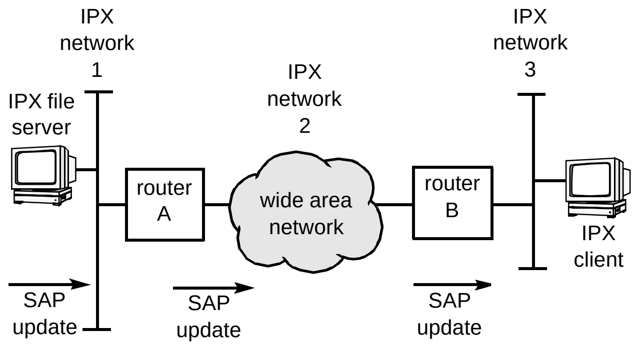

Fig 5 IPX file server SAP updates.

- ‘Keep-alive’ information

  Keep-alives are used to ensure that end-to-end session integrity is maintained (i.e. that both end systems taking part in a session are still connected). Keep-alives are used extensively in Novell NetWare environments [10], whereby the Novell NetWare Server generates periodic updates to maintain connectivity information. Keep-alives will either initiate or maintain a dial-up ISDN connection; this will further increase the total connection time.

  The types of keep-alives used in a Novell NetWare environment include:

  — ‘watchdog keep-alives’ which are sent by a NetWare Server to each attached client and are used to monitor the connection status of NetWare clients,

  — ‘sequenced packet exchange (SPX) keep-alives’, an SPX protocol, used by Novell NetWare to ensure guaranteed sequence and delivery of packets, which is employed in applications such as Novell's Remote Console (RCONSOLE) and Remote Printer (RPRINTER), and, when an SPX connection is established, both ends of the connection exchange keepalives to monitor connectivity,

— 'serialisation packet keep-alives', a Novell NetWare built-in protection scheme that transmits a packet containing a unique serialisation number between servers to protect licensed copies of NetWare from being reused on multiple servers.

These IPX keep-alives are inherent in Novell NetWare versions 3.x and 4.x. NetWare 4.x uses NetWare Directory Services (NDS) and this requires the use of a number of synchronisation protocols to ensure that directory and timing information is consistent across all servers. These additional synchronisation protocols add extra complexity when running NetWare 4.x across ISDN (particularly in the area of reducing ISDN call charges).

It should be noted that TCP/IP applications such as FTP (file transfer protocol) and Telnet do **not** make use of application layer or transport layer keep-alives to ensure that the session is alive, and therefore ISDN B-channels will only be raised when there is user data to send.

5.2 Administrative traffic across dial-up — solutions

There are many bad ways to design ISDN data networks! The administrative traffic exchanges discussed in the previous section combined with poor logical network design frequently typify this, with the result that unnecessary ISDN call charges are incurred. Fortunately, there are a number of techniques which can be used to reduce call charges and also maintain a reliable and stable network. These techniques include the following.

- Static routes

  As described earlier, dynamic routeing protocols send updates either periodically or on a 'triggered' basis. However, this will cause an ISDN link to be initiated (and hence incur call charges) each time a routeing update is to be sent, or will cause the ISDN B-channel to be enabled continuously.

  Static routes, however, can be used to reduce such call charges. They are configured by the network administrator and do not change unless the network administrator changes them. The configuration of static routes can be a significant administrative overhead in large networks. However, static routes do not cause the ISDN link to be established because there is no periodic forwarding of routeing updates. Certain router vendors have tried to combine the benefits of a dynamic and static routeing protocol for use across ISDN-based networks. One idea is 'snapshot routeing', which withholds topology change information until user traffic needs to be sent over the ISDN network; in this way routeing information is 'piggybacked' on to the user information. Other ideas use 'triggered RIP', in which RIP updates are only forwarded when there is a topology change; however, this may lead to oscillation of the ISDN link state for minor network changes.

- Static services

  Using a similar concept to static routes, static service configuration can be used in Novell NetWare networks — here the reachability of all services (such as file and print servers) are manually configured and advertised by the router. This reduces ISDN call charges (SAP updates are not forwarded periodically; they are blocked at the advertising end). However, configuring static services increases administrative overhead and bears the risk that a server not actually on-line is still being advertised at the spoofing end. Therefore clients will continue to attempt to attach to the server, with corresponding activations of the dial-up link.

- ISDN timers

  Once an ISDN B-channel has been initiated by some user data then there must be a mechanism for clearing down the call; the most common method is via the use of idle timers. Each user data packet will reset a timer; if there are no further user data packets then the timer will expire in due course and the ISDN B-channel will be disconnected. Disconnection of the ISDN B-channel will be transparent to the end user. Due to the fast call set-up times of ISDN (less than 1 second), when there is further user data to be transferred, the ISDN B-channel will be re-established and the users end-to-end session seemlessly maintained. There may be a further short delay if any security features are enabled (e.g. CHAP authentication).

  In addition to timers which control the duration of ISDN calls, certain vendors provide time-of-day control to make use of cheaper pricing structures. Note that the configuration of timer values should reflect a user's traffic profiles and also the tariffing structure, particularly where charging is based on a per-unit as opposed to per-second basis.

- Filtering

  In conjunction with routeing and timer configuration, the majority of router vendors also provide extensive filtering capabilities. These can be separated into two categories — filters configured to control which packets cause the ISDN data channel to be activated (and hence cause specific timers to be reset), and filters configured to control which packets are forwarded by the router.

Taking the former case, filters can be configured to restrict which users (e.g. managing director), protocols and applications (e.g. file transfers) can bring the ISDN link up. Most vendors allow filtering on source/ destination address, protocol type and also port number. These facilities enable tight control over which packets cause the ISDN link to be established; however once the link is established, the latter method of filtering must be used to control which packets are actually transmitted across the ISDN link. The majority of vendors provide filtering capability for most routeable and non-routeable protocols, e.g. filtering of Novell SAP packets and serialisation packets.

Filtering may also be used for security purposes, as discussed in section 6.

- Spoofing

Section 5.1 described how Novell NetWare environments make significant use of keep-alive service packets to maintain connectivity information. Watchdog and SPX keep-alives are used to ensure that connectivity between client and server is maintained. Spoofing is a technique which allows customer premises equipment to respond to keep-alive messages as if they were the client or server device, thus preventing keep-alive messages from initiating the ISDN link. Figure 6 depicts the use of spoofing for watchdog keep-alives; however, the same concept may also be applied to SPX spoofing.

Although spoofing will prevent the ISDN B-channel from being initiated unnecessarily, a limitation is that the NetWare server or device transmitting the keep-alive packet will continue to maintain the end-to-end session if a client machine is powered down without logging-off gracefully from the server. (A practical solution is to configure the server to clear down all sessions at a particular time of day; additionally, certain vendors provide the facility to automatically clear down sessions after a period of inactivity.)

As mentioned previously, Novell NetWare 4.x adds great complexity when clients and servers are located at different sites across ISDN, due to the synchronisation requirements of the database and timing protocols.

Although these protocols will cause the ISDN B-channel to be initiated, Novell and other vendors have defined filtering and spoofing methods which make the use of NetWare 4.x across ISDN more cost-effective.

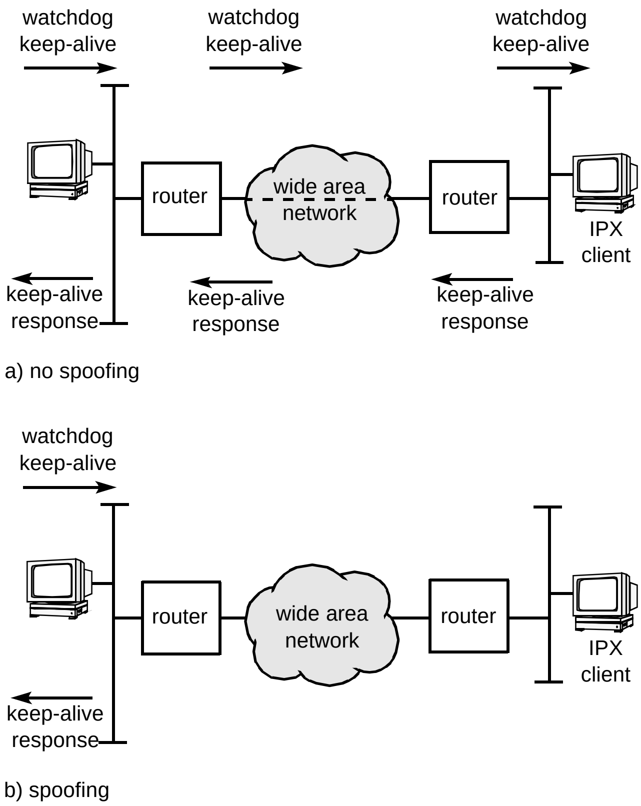

Fig 6 Spoofing of Novell NetWare keep-alives.

6. Security

Security is increasingly a key concern for any business connecting to a public network, and the explosive growth of the Internet has highlighted this. Security is necessary to safeguard sensitive data and protect against fraudulent use of corporate resources; authenticating users on the system and limiting authorisation rights are key to good security. Although perfect security does not exist, there are a number of techniques available which offer varying degrees of protection. It is common for a number of techniques to be collectively employed for increased protection. These are implemented in some form by all organisations to protect company-sensitive information.

Security techniques available for data networking using the ISDN include the following.

- CLI/sub-addressing

Calling line identity (CLI) may be used by the called station to filter connection requests — the router may be configured to accept calls from certain ISDN numbers only. Sub-addressing additionally enables users to address individual host machines which are collocated on a single ISDN S/T bus, and dialled using a single ISDN number. The user requires prior knowledge of the sub-addresses to access particular machines.

- User authentication

  All ISDN connections use the point-to-point protocol (PPP). PPP supports user authentication employing either PAP or CHAP [5]. BT recommends the use of CHAP wherever possible, which involves the called station (i.e. router) sending a challenge to the remote station, i.e. another router, which replies with a value produced by applying a one-way hash function to the 'challenge string' and other specifics, as required, (i.e. the 'password' is not transmitted across the link). If this reply matches the station's own hash calculation, authentication is acknowledged.

- Filters

  These may be applied to particular interfaces within a router, and may be used to filter users based on a layer 3 network address and protocol type (e.g. IP, IPX). A firewall is a comprehensive filter with additional, powerful facilities to detect faked packets. Both the source and destination addresses can be used for this purpose, and may, for example, prevent access to a particular server for any dial-in user. Network address substitution is straightforward if connecting via an ISDN point-to-point link, though the filters, used alone in this environment, represent only a very basic level of security.

- Dial-back

  On connection, the call is immediately cleared (usually following authentication) and the user is called back on a preconfigured number; it is possible to negotiate this within PPP. This represents a much higher degree of security, and has the added advantage that connection costs will be accumulated at a central site (typically the company host site). Dial-back is not recommended where short transfer of one or two packets is to take place, for example, with an SNMP poll to obtain basic router information.

- Authentication server

  An authentication server is a more recent development. It is a dedicated server for remote access log-in, and provides authentication, authorisation and accounting (AAA) functionality. Examples of authentication servers include the Terminal Access Controller Access Control System (TACACS) and Remote Authentication Dial In User Service (RADIUS). They are accessed by the called router when the initial ISDN connection is made. Users are authorised against details held within the server, which may, for example, be configured for dial-back, restrictions on command usage, restrictions on access time (e.g. daytime only) and connection period, all on an individual or group basis. Authentication servers also generally provide facilities for logging accounting information, including logging who has dialled in, when and for how long, thereby providing an effective audit trail. An authentication server may be considered to offer one of the highest degrees of security currently available to organisations supporting access to dial-in users.

- Encryption

  The aforementioned techniques aim to restrict access to a central dial-in site. However, thus far there is no protection for data while it is carried via the public network; this could potentially be illegally monitored or intercepted by a third party. Encryption is employed in order to ensure data cannot be understood by a third party as it requires a unique key, held only by the originator and receiver, in order to decrypt the information. There are various levels of encryption, the level of security offered being derived from the algorithm used, the key size and the frequency with which keys are changed. Disadvantages of encryption include the overhead placed on the router (although it is common to use dedicated encryption machines) and difficulties associated with key distribution. Encryption is common in financial and defence type applications.

Current solutions will typically use CLI, PPP with CHAP and access lists to provide basic security. In addition, there are usually security measures in place for access to local servers, e.g. a Novell NetWare mail server requires password protected log-in. There is a gradual migration to dedicated remote access servers as an increasing number of organisations become more security conscious, and demand at least part of the functionality offered by such dedicated servers.

More details on security in data networks can be found in Forrester et al [12].

7. Remote management

BT offers the management of ISDN-attached CPE, which includes the ability to carry out remote software upgrades to all attached devices. It also offers the ability to 'poll' the CPE periodically for various statistical information and also to receive triggered messages, or traps, from the associated CPE.

Internetworking vendors are constantly providing revisions and fixes to their software — vast savings can be made if software upgrades can be carried out remotely, as opposed to the costly and time-consuming business of sending field engineers to site. In addition, this form of upgrade can be carried out from one management centre, allowing expensive management hardware, software and expertise to be deployed centrally. This form of remote access must be secure (using secure protocols such as PPP CHAP and techniques such as ISDN call-back as discussed previously) and upgrade schedules must be agreed in advance with the customer.

The periodic 'polling' of CPE allows network managers to build up a protracted view of the performance of the CPE. This is made possible because all of the major internetworking vendors support the simple network management protocol (SNMP [13]) and provide SNMP agents on their equipment. SNMP assumes the use of TCP/IP and has the ability to gain a vast array of information from the agent via a management information base (MIB). The MIB provides a repository of information (typically including system, interface, IP, TCP information); however, as there is no industry standard ISDN MIB, certain vendors have implemented their own private MIBs to provide ISDN information (e.g. ISDN B-channel usage, maximum call duration, number of calls refused, etc). A possible use of SNMP is depicted in Fig 7.

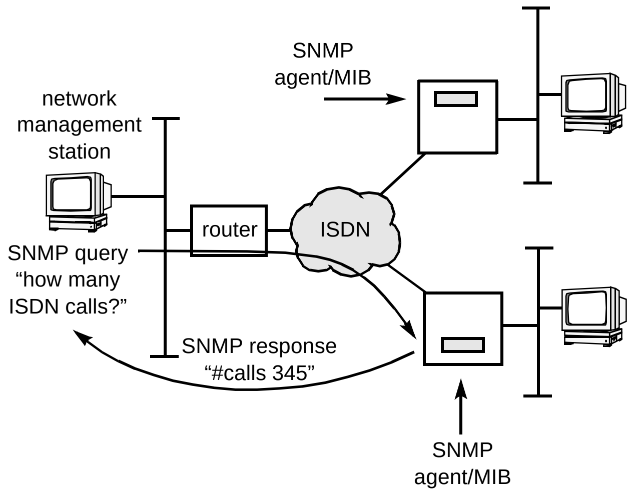

Fig 7 SNMP management over ISDN.

A key issue pertaining to the management of CPE across the ISDN is the reduction/elimination of call charges for management duties. Although call charges cannot be avoided when upgrading software, a number of techniques are used to check that an ISDN-attached device is still operational (without sending SNMP queries) and without incurring call charges. Over recent years, out-sourcing of network management and ownership has greatly increased, so it is important that CPE (supporting all service types) can be managed remotely.

8. Case study

The following information is based on the provisioning of a real customer ISDN network, and gives an example of the possible network requirements for 'Conversion Building Society' (CBS) (see Fig 8).

The building society sites consist of a central host site, small, medium and large branch sites and a legacy authentication server. In addition, teleworking is supported for members of the board who are frequently away on business or working from home. The ISDN network operator provides network management services from the central network management centre (NMC); this management is 'reactive' only (no pro-active polling of the routers). Each of the site types, their traffic demands, and consequent network requirements are now discussed in turn.

8.1 Teleworker

In this example, the teleworker requires occasional access to the host site, potentially from anywhere in the world. Typically this is to access e-mail and for file transfers. Connection to ISDN is therefore via a simple terminal adapter (TA), which connects to the teleworker's computer via a serial interface (e.g. V.24). The PPP protocol runs between the TA and the far-end (host site) router. The TA may be an interface card within the PC itself, thus negating the need for a serial connection. There is therefore no intermediate router at the teleworker's location. Only a single B-channel is ever used and authentication is performed using the server at the host site end. Dial-on-demand is therefore the ideal service for the teleworker.

8.2 Network management

Management access requires ISDN connection. This generally requires a Telnet-type interactive session to the router concerned, and connectivity from the NMC to the ISDN is via a single BRI only. This method may also be used to remotely upgrade the software within the customer's routers.

8.3 Small branch site

The majority of CBS sites are of this type, and contain a single ISDN-attached router. Traffic demands are low and tend to peak towards the end of the working day. A single ISDN B-channel at 64 kbit/s is adequate and is therefore used for the majority of connections, which are typically of one minute duration with low rates of data exchanged. However, towards the end of the working day, traffic demand reaches a peak and a second B-channel is usually initiated to form a multilink PPP group. Initiation of this additional channel is automatic and based on configured load threshold levels on the first B-channel. Authentication of the branch takes place at the host site using the calling line ID and authentication server. All IPX keepalives are spoofed by the router, and routeing is statically configured — there is no dynamic routeing protocol enabled. The Novell NetWare server at the host site is statically configured on the branch site routers and thus advertised by them.

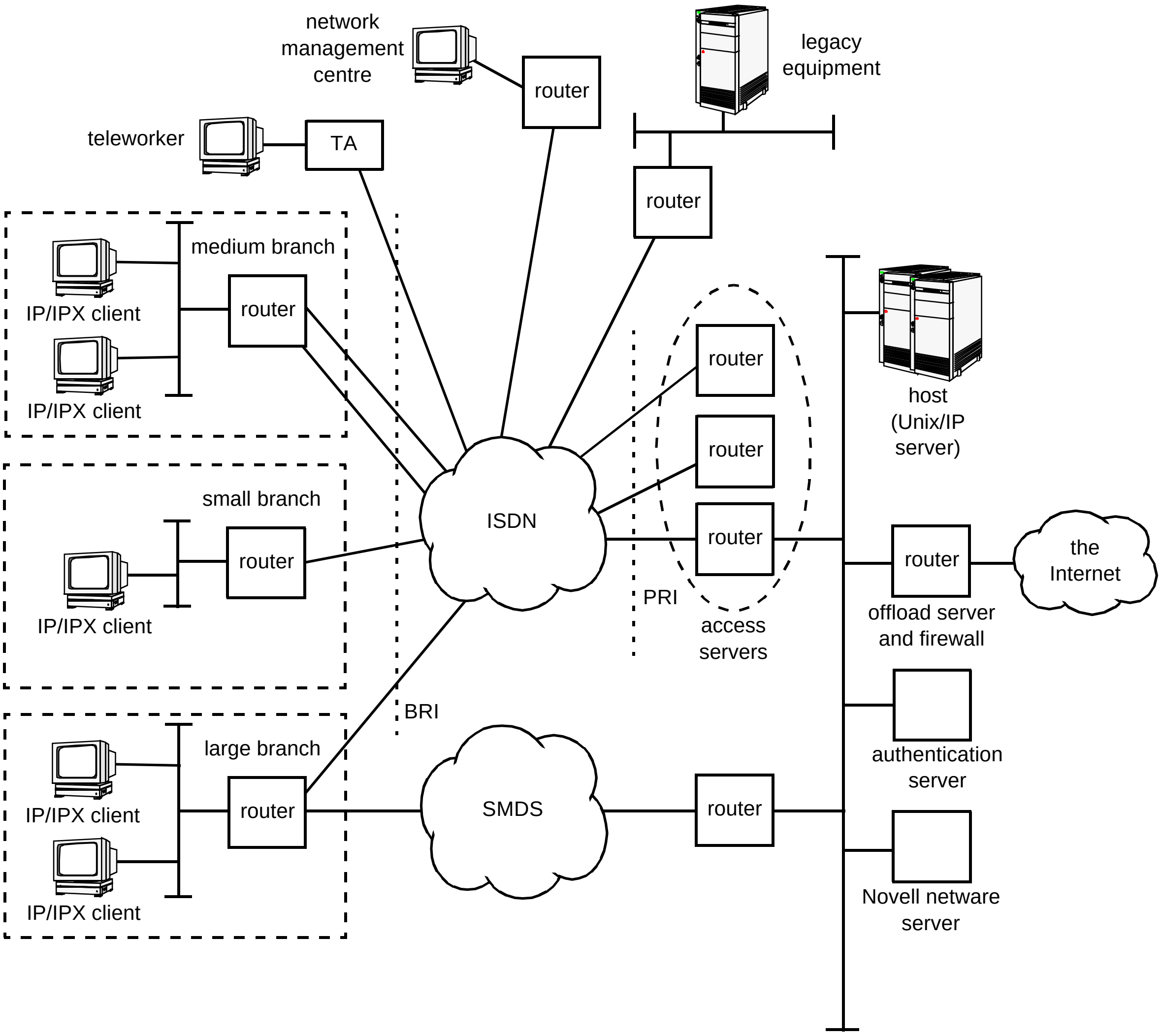

Fig 8 'Conversion Building Society' network.

8.4 Medium branch site

These sites are virtually identical to the small sites, other than higher overall traffic demands. It was therefore recently decided to connect these to the ISDN using two BRIs, thus a 256 kbit/s connection may be made if necessary. Bandwidth-on-demand has therefore been implemented. The routers at each medium-sized site have a higher processing capability than those at the small branch site, and to maximise link utilisation compression has also been enabled. This does not impact on any other router tasks due to the higher processing capabilities. All IPX keepalives are spoofed by the router, and routeing is statically configured — there is no dynamic routeing protocol enabled. The Novell NetWare server at the host site is statically configured on the branch site routers and thus advertised by them.

8.5 Large branch site

Traffic demands from the large branch sites are significantly higher than those from other sites. For this reason a dial-on-demand service did not make sound economic sense, and these sites have been connected to the host site via BT's SMDS service [14]. Due to the critical data that these sites process, it is imperative that connectivity to the host site be available at all times; for this reason each large site has a single BRI connection for back-up. In the unlikely event of primary network (SMDS) failure, connection will be made via the ISDN. This represents a significant reduction in bandwidth available, and for this reason both compression and priority queuing have been enabled for any traffic carried via the ISDN. IPX keepalives and services are **not** spoofed as there is no advantage in doing so, due to the primary network being SMDS.

8.6 Host site

There are a number of ISDN-attached access servers (routers) at the host site, which are all connected to the ISDN via the same hunt group, using primary rate interfaces. Potentially, the host site may support many tens of separate connections, although there may be multiple B-channels belonging to particular host sessions. Multichassis Multilink PPP has therefore been enabled on these routers, and a separate off-load server is used to minimise the processing overhead on these routers. The backbone Ethernet connects these to the host, and access to the Internet is possible via a firewall router. The firewall prevents unwanted users accessing CBS's resources, and uses filters to prevent access to the Internet from any of the branch sites; access is only permitted from the host site. There is no requirement for the branch sites to have Internet access, and the board of directors has explicitly requested that it be prevented. IPX keepalives for small and medium branch sites are spoofed, and all routeing is statically configured.

8.7 Support of existing systems

As CBS has evolved to electronic communication, and this in itself has evolved, there has been a number of legacy systems still requiring support. One of these is an authentication server used by one of the mission-critical applications running on the host. The server was intentionally located at a different site, for physical diversity. Communication between the host and server totals less than an hour a day, and volumes of traffic are low. However, due to the sensitive nature of the data CBS did not wish to transfer this information in plain text format across a public network; encryption is used. Failure of communication between the host and server is not considered catastrophic by the customer, hence reliance on ISDN connectivity only. Spoofing and static routeing are again configured on the router.

CBS is actively monitoring network traffic, and predicts that traffic at the medium sites will reach levels similar to those currently experienced at the large sites, within five years. It is expected these sites will be migrated to SMDS connectivity in due course, with ISDN connectivity for back-up only.

9. Conclusions

ISDN holds a unique position within the data networking world. It offers scalable on-demand digital connectivity over the wide area, and supports a wide range of networking applications. These may be summarised as:

- dial-in (teleworking),
- as a core network technology,
- as back-up (an alternative core network technology),
- for the remote management of routers.

Currently, ISDN is primarily used as a back-up mechanism for connectivity in the unlikely event of core network failure, but there is a significant trend towards greater use of ISDN as a primary network.

There has been considerable effort in recent years to enhance the functionality available within routers to support transport of the most prevalent protocols across the ISDN. The generic aims of this development remain to maximise bandwidth utilisation and minimise connection times. However, issues such as security are becoming increasingly important and there have been significant developments in recent times. A wide range of techniques are now available in order to optimise ISDN network design, both in terms of minimising connection time and maximising support of the more prevalent protocols in use.

Further developments will occur as the ISDN evolves, in particular with functionality implemented in ISDN-attached routers. Its unique role within the data networking world will ensure that the current rise in demand for ISDN continues into the foreseeable future.

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The following is a useful Web-based index search form for RFCs:

http://www.nexor.com/public/rfc/index/rfc.html

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